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Applied Data Mining
Assignment 5 Project Update

Taxi Cancellation Project

This project update continues the taxi cancellation prediction project from Assignment 4, but the main focus now is improving the project so it can become a stronger final business analytics report. In Assignment 4, I already created the foundation of the project by using the taxi-cancellation-case.csv dataset and the case5.R script to predict whether a taxi booking would be cancelled. The project included data cleaning, factor conversion, date-based feature engineering, trip length calculation, model training, and model evaluation. The models tested included K-Nearest Neighbors, Decision Tree, Boosting, and Logistic Regression.

For the next stage of the project, I identified that the original R script needs to be improved before it can fully support the final report. The previous script was useful for testing the models, but it did not clearly print organized outputs when the full file was run. Also, the prediction object was repeatedly overwritten using the same name, pred, for each model. This made the evaluation section confusing because the final confusion matrix and gains chart could end up representing only the last model that was run instead of evaluating each model separately. For the final project, each model needs to have its own prediction object, such as pred_knn, pred_tree, pred_boost, and pred_logit, so that the models can be compared clearly and fairly.

Another important improvement is that the final R script should produce a clear data summary before modeling. This includes the number of rows and columns, the distribution of the target variable Car_Cancellation, and the percentage of cancelled versus non-cancelled bookings. This is important because one of the main findings of the project is that the dataset is highly imbalanced. Most bookings were not cancelled, which means that accuracy alone can be misleading. A model can appear to perform well simply by predicting the majority class, even if it does not successfully identify actual cancellations.

The updated version of the project will therefore focus on creating a cleaner final R script that prints organized results for each model, creates separate confusion matrices and gains charts, and produces a final model comparison table. This will make the final report stronger because it will not only show that the models were run, but also explain what the results mean from a business perspective. The final project will focus on 2 to 3 main analytic techniques, especially Decision Tree, Boosting, and Logistic Regression or KNN. Decision Tree is useful because it is interpretable, Boosting is important because it improves predictive performance by combining multiple weak learners, and Logistic Regression can serve as a traditional baseline model while also showing the limitations caused by factor-level issues and instability.

The goal for the final project is to turn the Assignment 4 modeling work into a polished business analytics report. Instead of only presenting code and screenshots, the final report will tell a complete story: the business problem, the structure of the data, the data preparation steps, the models used, the results, the limitations, and the implications for taxi companies. The key business insight is that taxi companies should not only focus on overall accuracy, but also on whether the model can detect actual cancellations. Predicting cancellations more effectively could help improve driver allocation, reduce wasted trips, minimize revenue loss, and support better operational planning.